

MGT 473: Data Science with Artificial Intelligence

Class: FT MBA

Term: Spring 2026

Time: 9:00AM – 11:50AM

Location: TBD

Slack: #TBD

Textbook (optional): [Homage to R](#)



Prerequisites: Required: None

Professor: Robert Evan Sanders

Email: rsanders@radly.ucsd.edu

Office Hours: By appointment. The best way to reach me is through Slack, either by DMing me or posting to the class channel. I respond to Slack faster than I respond to email!

Teaching assistant (TA): TBD

TA email: TBD

Office hours: TBD

R/Rstudio session: TBD

You don't need to be a coder to become a data scientist—you just need to know how to boss one around. In MGT 473, you'll learn to manage AI the way a CEO manages a team: set the vision, give the orders, and verify the results. Starting from zero coding experience, you'll go from writing your first lines of R to directing a full-blown AI data scientist that scrapes live data, runs causal inference, and builds production-ready analyses—all from plain English instructions. By the end of the quarter, you won't just be using AI; you'll be building custom AI agents that manage *other people* for you. The class replaces exams with weekly quizzes, runs in-class competitions with a battle-honors leaderboard, and uses business cases from companies like Alibaba and Coca-Cola. If you've ever felt like data skills are a superpower locked behind a wall of syntax, this course hands you the key. Business majors, this is your moment: the game has shifted from who can code the fastest to who can think the clearest, and that's you.

Artificial intelligence (AI) has significantly reshaped analytics. Today's most valuable quantitative analysts are not distinguished by coding-syntax mastery and cleverness of execution, but by their ability to quickly extract meaningful insights from data. What employers now prize is logical reasoning: the skill to direct AI agents effectively, while ensuring your agents are indeed executing what you instructed them to as you meant them to—that is, verifying “alignment” with your intentions. AI has also unlocked entirely new kinds of analysis that were heretofore prohibitively

costly. In a world with AI, analysts function more as architects and supervisors than as code craftspeople.

This course addresses both dimensions: how to work with AI agents to augment traditional data-science workflows, and how to explore entirely new frontiers of analysis that these revolutionary tools have made accessible.

OVERVIEW

Large language models (LLMs) such as Claude are powerful. To maximize your productivity—and stay relevant—you should aim to delegate as much of your workflow to these AI tools as you safely can. These tools are a great equalizer: with the right AI, an inexperienced but thoughtful data scientist can accomplish almost anything that an experienced data scientist can. Business majors now stand out as the finest data scientists since the critical analytics differentiator has shifted from technical expertise to generating useful insights.

This class will make you a more effective user of generative AI—especially in the context of data analytics. To best use generative AI, we must make one concession: we must transition from spreadsheets to computer code. Indeed, LLMs are far smarter when they think in terms of a programming language, such as R, than when they think in terms of Excel. An Excel framing hobbles an LLM because a language model can't be trained on ephemeral mouse clicks. There's a large corpus of R code online but no corresponding corpus of Excel cursor movements. Accordingly, a large *language* model can more deftly manipulate the syntax of a computer *language* than the options of a dropdown menu.

In this class, you will learn “natural language programming,” using AI to translate from the English language to the R statistical programming language. Beginning with foundational R skills (no prior coding experience required), you will delegate progressively more responsibility to AI agents over the course of the term. By the end of the class, you will be managing a full-fledged AI data scientist.

Rather than write computer code from scratch, however, you will learn how to “vibe code,” (1) learning and articulating the desired data transformations in English, (2) using AI to translate the spoken language into a computer language (specifically R), and (3) coming up with tests for the AI to demonstrate alignment and accuracy. You will soon be more productive programming in English than mousing in Excel. (We will confirm this with an in-class demo.)

In years past, this class has been taught where students were allowed to code with Chat-GPT from the outset. However, this proved to be a mistake, as it's difficult to learn the basics when AI pre-emptively answers you before you have time to think. Accordingly, we will introduce AI into our workflow more deliberately. Specifically, the class will comprise four chapters, each delegating progressively more responsibility to the AI.

Chapter 1 is titled “Understanding the AI.” This chapter will teach you the R statistical programming language—the lingua franca of data science. Just as Latin was indispensable to Renaissance scientists, proficiency in a statistical programming language, such as R, is indispensable to modern data scientists. LLMs can translate effortlessly between English and R. One might assume this diminishes the value of learning R—but the reality is precisely the opposite: LLMs make R knowledge more valuable. Before AI, learning R didn’t pay off until you could analyze data more effectively with code than with Excel, a high bar. But with AI, you don’t need to write code to leverage your R knowledge—you need only to read it. And you don’t even need to grasp everything, as 10% of R covers 90% of data tasks. Committing this crucial 10% to memory will pay off in the long run, just as it paid off to memorize the alphabet and the times tables. Memorizing the key R functions will enable you to quickly parse and verify sophisticated AI-generated code. In short, after this chapter, you will be able to understand your AI data scientist, which is the first step to managing it. We’ll learn the basics of R from a new textbook (written by my co-author, at Northwestern University). Zero programming experience will be expected—we will start from the ground up.

Chapter 2 is titled “Collaborating with the AI.” This chapter will introduce AI your workflow. You will learn techniques for natural language programming and for assigning empirical tasks to an AI. You will also learn how to learn from an AI. You will partner with the AI to derive insights from real business data.

Chapter 3 is titled “Managing the AI.” This chapter will hone your managerial vision. Whereas chapter 2 focuses on execution—you will be told what to do, and you will have to figure out how to do it—chapter 3 focuses on leadership—you will have to figure out both what to do and how to do it. This chapter will test both your design skills and your managerial skills: you will have to conceptualize the blueprints of your data analysis, and you will have to instruct AI agents to realize your plans.

Chapter 4 is titled “Promoting the AI.” This chapter will add another layer to the hierarchy, upgrading your AI from a data scientist to a middle manager. You will learn how to create AIs that extend your managerial scope. Specifically, you will create a custom GPT to oversee other students in the class. You will exploit AI’s extraordinary communication skills: rather than micro-manage your underlings via meetings and memos, you will embed your directives in a custom GPT, which will oversee the implementation details.

OBJECTIVES

Having successfully passed this course, students should be capable of:

- Writing basic R code for data analysis, without AI assistance—the best way to learn how to program with AI is to learn how to program without AI.
- Using AI in R for data analytics: from generating/debugging code to summarizing textual data.
 - Using AI to conduct large data analyses, such as scraping and analyzing social media data using an API to measure real-time consumer sentiment.

- Implementing fundamental causal inference strategies, such as difference-in-differences or A/B testing, to conduct counterfactual analyses (i.e. "what if studies") to glean the impact of business or policy interventions.
- Vibe coding, translating English instructions into production-ready R code. That is, prompting AIs and verifying output for the accurate results.
- Integrating these skills in a group project: design a question, gather data, apply causal tools, and present actionable insights.
- Extending your managerial scope with AI agents.
- Designing custom-made AI experiences.
- (Optional: Working with the frontier, high-powered AI integrated IDE, Microsoft Visual Studio (VS) Code + Cursor, Claude Code.)

COMPUTING

We will be using two main tools in this class: R for data analysis and RStudio as our programming environment.

About R. R is a free, open-source statistical software package that is used by most firms on the frontier of data science and business analytics. Unlike point-and-click platforms (e.g., SPSS, Excel, Minitab), R has a command line interface that gives the user complete flexibility when carrying out statistical analyses. This class does not presume that you have any prior experience using R, but it *does* presume you have some basic familiarity with analytics (e.g., an introductory statistics class) and with logic-based programming.

Rstudio. RStudio is a free, powerful integrated development environment that transforms how you write and organize code. It brings together your scripts, output, visualizations, and files into one clean interface—making it far easier to stay organized, catch errors, and iterate quickly compared to running code in a basic console.

Software Installation:

Download the most recent version of R from: <https://cran.r-project.org/>

Download RStudio following the instructions from: <https://posit.co/downloads/>

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COURSE SCHEDULE

	Preface Get set up with R and RStudio <u>Assignments:</u> Start hw_intro.pdf Take introduction survey
Mar. 30	Class 1 Chapter 1: Understanding the AI <u>Topics:</u> Cover slides_introduction_and_tibble_1_student.Rmd Course overview Data-science and AI overview Learn the general empirical workflow – Variables – Vectors – Logicals, doubles, and integers Learn how to represent data Start hw_quiz_prep_1.pdf (I) <u>Assignments:</u> hw_intro.pdf (I) due before class introduction survey (I) due before class <u>Textbook:</u> Preface + Chapters 1.1–1.2.4
Apr. 6	Class 2 <u>Topics:</u> Take quiz on hw_quiz_prep_1.pdf (I) Cover slides_tibble_2_and_quartet_1_student.Rmd – Character strings, dates, and factors – Tibbles

	– Projects – Learn how to structure data – Calling functions – Conditional statements – Functional assembly lines – Learn how to manipulate data Start hw_quiz_prep_2.pdf (I) Start hw_automate.pdf (I) (when finished, upload to Slack) Assignments: hw_quiz_prep_1.pdf (I) due before class Textbook: Chapters 1.2.5–1.2.6 + 2.1–2.2.3
Apr. 13	Class 3 Topics: Take quiz on hw_quiz_prep_2.pdf (I) Cover slides_quartet_2_student.Rmd – Formatting code – mutate() – summarise() – group_by() – Battlefield maneuvers, round 1 Learn how to distill data Start hw_quiz_prep_3.pdf (I) Assignments: hw_quiz_prep_2.pdf (I) due before class Textbook: Chapters 2.2.4–2.2.9
Apr. 20	Class 4 Topics: Take quiz on hw_quiz_prep_3.pdf (I) Cover slides_quartet_3_student.Rmd – filter() – group_by() filter() – Group .by

	– Battlefield maneuvers, round 2 – Judges case Learn how to filter data Finish hw_automate.pdf (I) Start hw_quiz_prep_4.pdf (I) Start hw_judges.pdf (G) <u>Textbook:</u> Chapters 2.2.10–2.2.11 <u>Assignments:</u> hw_quiz_prep_3.pdf (I) due before class
Apr. 27	Class 5 Chapter 2: Collaborating with the AI <u>Topics:</u> Take quiz on hw_quiz_prep_4.pdf (I) Cover slides_vibe_student.Rmd – Battlefield maneuvers – Give in to the Vibes – Vibe Coding in Data Science – Experiment (R vs Excel comparison) Continue hw_judges.pdf (G) <u>Assignments:</u> hw_quiz_prep_4.pdf (I) due before class; Chapter 2.3 (Lab: Italian Judiciary) (Optional: Listen to “The Future of Empirical Research in the Age of AI” from <i>Not Another Politics Podcast</i>) before class
May 4	Class 6 <u>Topics:</u> Take quiz on hw_judges.pdf (I) Cover slides_join_eversight_student.Rmd

	– Are you a better dealmaker than AI? – Joining Data – Battle Honors Start hw_oversight_1.pdf <u>Assignments:</u> hw_judges.pdf (G) due before class hw_oversight_1.pdf (G) due by May 6 end of day <u>Textbook:</u> Chapters 4.2.6–4.2.7 + Chapter 4.3 (Lab: Oversight)
May 11	Class 7 Chapter 3: Managing the AI <u>Topics:</u> Take quiz (I) on hw_oversight_1.pdf / hw_oversight_2.pdf Cover slides_punchout_orchestrate_student.Rmd Start hw_coke.pdf <u>Assignments:</u> hw_oversight_2.pdf (G) due before class <u>Textbook:</u>
May 18	Class 8 <u>Topics:</u> Take quiz on hw_coke.pdf Cover slides_alibaba_student.Rmd – AI management – Data visualization – Learn how to create rapid AI prototypes – Battle Honors Ali Baba

	Start hw_final_1.pdf Start hw_final_2.pdf <u>Assignments:</u> hw_coke.pdf due before class <u>Textbook:</u> Chapters 3.3 (Lab: Alibaba)
May 25	Class 9 Chapter 4: Promoting the AI <u>Topics:</u> Take quiz on hw_final_1.pdf Take quiz on hw_final_2.pdf Cover case_888.pdf Cover case_505.pdf Finish hw_final_1.pdf Finish hw_final_2.pdf <u>Assignments:</u> Read hw_final_1.pdf due before class Read hw_final_2.pdf due before class <u>Textbook:</u>
Jun. 1	Class 10 <u>Topics:</u> Start hw_final_4.pdf Finish hw_final_4.pdf <u>Assignments:</u> Finish hw_final_3.pdf due before class
[Post-term] Jun 5	<u>Assignments:</u> hw_final_5.pdf due

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DELIVERABLES

- **(I)** – Independent, individual work only. No collaboration or consultation allowed.
- **(G)** – Students may work together in groups and turn in one deliverable for the entire group.

LATE SUBMISSIONS:

Deadlines are strict. Barring some kind of emergency, *if you fail to email the TA in advance, there is a 50%-point reduction the first day and a 75% reduction the second day. After the third day, there will be no points awarded.*

PARTICIPATION (I):

Active participation is a component of the class experience. Class attendance, class engagement, Slack-channel activity, and your performance on the in-class activities will determine your participation grade. Attendance is a necessary but not sufficient condition for participation. Switching sessions is allowed, but please notify me in advance for that specific week.

You may miss one class, no questions asked, provided you tell the TA. This “registered” absence is to account for possible self or family illness, emergencies, immovable job interviews, and the like. You must email the TA at least 12 hours before class to register that you are missing class. A registered absence will result in an “NA” participation score for that week, which means I will overweight the other weeks to determine your final participation score. It will also result in a score of zero on the quiz, however, I will drop your lowest quiz at the end of the quarter. Nonregistered absences will reduce your class participation score, and you will get a score of zero on the quiz.

Battle honors: Almost every class features an in-class game or competition. Winners earn battle honors, which are recorded on a class leaderboard and contribute to your participation grade. Most battle honors follow a two-step process: first, a Challenge Question appears in the slides, and you post your answer on Slack by the end of class. If you answer correctly, you become eligible for the Battle Honors Question, where a randomly selected student from the eligible pool must explain what a piece of code does in a single sentence. Get it right, and the honor is yours. Your battle honors are displayed as a functional assembly line—a chain of honor names linked with %>% operators, just like R code—and your score is the total character length of your chain. A leaderboard will be posted at the midpoint of the quarter, and an awards ceremony will be held during the final class.

HOMEWORK (I, unless otherwise specified):

Homework assignments will be posted online prior to class sessions and are to be completed individually unless otherwise specified. Some assignments are part of case studies that span multiple homework sets. For case-based assignments, the typical sequence is as follows:

1. Initial assignment: Reading and analyzing the case
2. In-class assessment of case comprehension quiz
3. Follow-up assignment: Laboratory work based on the case

All assignment due dates and submission requirements will be clearly indicated on the course website.

If the homework gives the option to work in a group of 2, please sign up for a group using the following links:

[To form groups](#) and [submit assignments](#), please follow the hyperlinks for instructions.

QUIZZES – (I)

This course has 8 quizzes, which collectively replace the traditional midterm and final exams. These quizzes break your assessments into smaller pieces throughout the term, so this class does not include a separate midterm or final exam.

AI POLICY

Quizzes are pen-and-paper only—closed-book and closed-AI—to promote learning and retention. For all other assignments and activities, you are encouraged (and expected) to use AI tools.

GRADING

Evaluation on:	Points [or percentage]
1. Class Participation	10%
2. Homework	40%
3. Quiz	50%

This is an elective course, and therefore there is no predetermined grading distribution. Therefore, the end goal should be to learn the concepts thoroughly and how to think like a data scientist.

Regrades:

If there is a very simple mathematical error or something I missed, you can bring it to me immediately after class or email the TA. Any other errors need to follow the following procedure: begin the following procedures for a regrade:

1. Within 14 days, email the TA explaining in a high level of detail each issue and why you believe the grade was incorrect.
2. Please email me to setup a face-to-face meeting in which we can discuss the email.
3. Note that **there is a possibility that your grade decreases in a regrade**, as I will have the TA regrade the work in its entirety.

COURSE POLICIES

Professionalism. By enrolling in this course, you are taking part in a small community that has come together for the purpose of learning. This entails two forms of professionalism that are expected in the classroom. First, you are expected to behave with the professionalism you would bring to the workplace. Behavior that would not be appropriate in a team meeting at work is probably not appropriate for class either. Second, you are expected to approach the class with the goal of actively maximizing both your learning and the learning of your classmates. Based on these principles, a few specific pointers of common problems to avoid:

- Please come to class on time, and, more importantly, please try not to walk in and out of class. Walking in and out of class is more disruptive than you realize!
- Please do not view videos/games/social media on your laptops in manner distracting to other students. While the amount of effort and attention you put into your own education is up to you, you don't have the right to infringe on other students who are attempting to learn.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/manual/appendices/appendix2.pdf>

How the Honor Code applies to this course:

This means, when an assignment is denoted "I," you may not communicate about the assignment with your peers in any form. Additionally, group assignments may be discussed within-group, not across-groups. Sharing classroom materials with students outside of the class is also not permitted. Any violation of this policy constitutes a breach of the honor code and academic integrity. Students may use

tutoring, but the above still applies. If you are uncertain about the fair scope of tutors, please email me for clarification.

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or fosorio@ucsd.edu.